High Performance Spaceflight Computing

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For decades, NASA has advanced on-board spacecraft computer processors that coordinate and execute the functions needed to support mission success.

Space computing originated in the 1960s with the Apollo Guidance Computers, which were pivotal for guidance, navigation, and control computations during NASA's first Moon missions. For decades, radiation-hardened processors have been the backbone of the agency's space exploration missions.

NASA has landed computers on other planets and operated them for years in extreme conditions, as demonstrated by the Mars rovers. These computer processors have also powered several NASA orbiters, capsules, and space telescopes.

While legacy processors have enabled some of NASA's greatest achievements, the next generation of space missions will increase in complexity and length, which will benefit from greater computing power, autonomy, and resilience. To meet the needs of this challenge, NASA and industry leader Microchip Technology Inc. entered a public, private partnership combining agency and commercial investments to develop a new solution: High-Performance Spaceflight Computing.

The High-Performance Spaceflight Computing project is a next-generation system-on-chip that delivers over 100 times the computing capability of current space processors. By integrating computing and networking into a single device, this technology significantly reduces system cost and power consumption. Its scalable architecture allows unused functions to power down, optimizing energy efficiency for critical operations.

The High-Performance Spaceflight Computing family of processors includes multiple distinct but compatible technologies for scalable mission needs. The radiation-hardened version of the processor is built for geosynchronous, deep-space, and long-duration missions to the Moon, Mars, and beyond, capable of operating in harsh environments while supporting real-time autonomous tasks. Tailored for the commercial space sector, the radiation-tolerant version of the processor provides fault tolerance and cybersecurity for low Earth orbit satellites.

Using advanced Ethernet to connect multiple sensors or cluster several chips, High-Performance Spaceflight Computing technology allows spacecraft to process massive amounts of data onboard and autonomously make real-time decisions, such as driving rovers at high speeds or filtering scientific images. Continuous system health monitoring and an integrated security controller ensure these complex operations remain safe and reliable.

The High-Performance Spaceflight Computing technology is a nationwide, public-private development effort anchored by NASA, Microchip, and a broad ecosystem of academic and industry partners. This collaboration reinforces U.S. leadership in spaceflight computing, strengthens supply chain resilience and security, stimulates regional economies, and drives innovation and high-tech workforce development across the nation.

This new technology has the potential for use on all future space missions, but unlike traditional space-specific chips, High-Performance Spaceflight Computing has a design platform for other Earth-based uses.

Adopting the same high-performance computing, network switching, high-reliability and cybersecurity technologies, the company's processors enable mission-critical edge computing for Earth-based industries such as automotive, aviation, consumer electronics, industrial systems, and aerospace. These potential applications include drones, energy grids, medical equipment, communication services, artificial intelligence, and data transmission.

By leveraging a common technology base across space and terrestrial markets, High-Performance Spaceflight Computing helps strengthen domestic industrial capabilities and reduce risk and cost for both government and commercial users.

The Space Technology Mission Directorate's Game Changing Development program based at NASA's Langley Research Center in Hampton, Virginia, and NASA's Jet Propulsion Laboratory led the end-to-end maturation of NASA's High-Performance Spaceflight Computing by developing mission requirements, funding competitive industry studies, selecting and contracting with Microchip, and guiding the project through design reviews and the project life cycle to delivery.

To learn more about these chips, visit:

https://go.nasa.gov/4cIGUKu

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